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Paleomagnetism of the Lower Triassic Moenkopi Formation

ABSTRACT

Paleomagnetic studies on the upper half of the Moenkopi Formation have identified three reversed and three normal polarity zones; this brings the total number of polarity zones observed in this formation to ten. The variable thickness and presumed duration of these polarity zones when combined with smaller variations internal to individual polarity zones suggest that this reversal sequence should be very useful as a regional correlation tool.

Several zones of anomalous magnetization have been recognized within the formation. These anomalous zones constitute less than 5 percent of the total sample population but are very important, because they may assist in the recognition of individual polarity zones.

The pole position derived from this study is at 103° E., 55° N. (dp 1.6°, dm 3.2°) and, due to the large number of samples and the length of time represented, probably represents an excellent average for the Lower Triassic of North America.

INTRODUCTION

Observations of both normal and reversed magnetic polarity have been made on rocks of all ages, yet for rocks older than the late Cenozoic, the time sequence of these reversals is virtually unknown. Two long periods of constant polarity have been identified, one in the Permian (Irving and Parry, 1963) and one in the Cretaceous (Helsley and Steiner, 1969). Other than these periods of constant polarity, no definitive polarity sequences are recognized generally, although several attempts have been made to use locally developed reversal sequences as an aid in stratigraphic correlation (Khramov, 1958; Picard, 1964; Burek, 1967; Helsley, 1969). The purpose of this paper is to extend our knowledge of the polarity-reversal sequence for Lower Triassic rocks of the Colorado Plateau so that the polarity reversal sequence can be used in the future in regional correlation of these otherwise unfossiliferous rocks.

PREVIOUS WORK

Studies of Triassic rocks from the Colorado Plateau (Runcorn, 1956; Kintzinger, 1957; Collinson and Runcorn, 1960) have shown that both normal and reversed polarities are characteristically present, although no attempt was made to define the polarity sequence. Helsley (1969) has defined a polarity-reversal sequence for the lower half of the Lower Triassic Moenkopi Formation and has done reconnaissance sampling of the upper portion of the Moenkopi Formation. He showed that these rocks are remarkably stable and that thermal demagnetization was effective in removing any secondary component that may have been present. The magnetization of the Moenkopi Formation has been shown to have been acquired penecontemporaneously with deposition (Baag and Helsley, 1974).

GEOLOGIC SETTING

The Moenkopi Formation crops out throughout the Colorado Plateau region. Its regional stratigraphy and history are discussed by McKee (1954) and more recently by Blakey (1973). In the western portions of its outcrop region, the Moenkopi Formation is characterized by a marine sequence including gray shales, carbonates, and evaporites in which red to maroon shales are interbedded. To the east, the formation becomes more clastic, and in eastern Utah and western Colorado, it consists of brown and red-brown sandstone, siltstone, and shale with minor amounts of gypsum and a few locally developed conglomerate lenses.

The region sampled for these studies is in the thickest and most continuously deposited portion of the clastic facies of the Moenkopi Formation south of Gateway, Colorado (Fig. 1), in an area in which the geology has been described by Shoemaker (1955). Bedding within the Moenkopi Formation is generally even and continuous, which greatly facilitates local lithologic correlations. In the area of this study, the formation consists of about 240 m of interbedded reddish-brown sandstone, siltstone, and shale with some interbedded conglomerate. The formation has been subdivided into four members on the basis of lithology by Shoemaker and Newman (1959). These members provide a means of local correlation between the outcrops sampled along the Dolores River and those sampled along Salt Creek (Fig. 1). These members are (from bottom to top): the Tenderfoot Member, a sequence of massively bedded bimodal sandstone, regionally equivalent to the Hoskinnini Member (Stewart, 1959); the Ali Baba Member, a sequence of interbedded sandstone, siltstone, and conglomerate; the Sewemup Member, a sequence of fine-grained sandstone, siltstone, and shale; and the Pariott Member, a sequence of multicolored siltstone, sandstone, and conglomerate. The lower boundary of the Sewemup Member is a distinct lithologic boundary in this area and has been used as a basis for stratigraphic correlation between adjacent sampling sections.

The age of the Moenkopi Formation of this region is well established as Lower Triassic (Dane, 1935; Shoemaker and Newman, 1959); however, the range of time represented by the formation is poorly known. On the basis of regional correlation with other strata of Permian and Triassic age, Shoemaker and Newman (1959) conclude that the lowest member of the Moenkopi may be either Permian or Triassic and that it is likely that the boundary lies within this member. The uppermost units of the Moenkopi Formation are considered by the same authors to be perhaps as young as Middle Triassic. Thus, one can only conclude that the rocks sampled for this study were deposited during an unknown period of time between uppermost Permian to lowermost Middle Triassic. Inasmuch as no major diastems have been noticed in this section, one can assume that sedimentation was essentially continuous.

Helsley's 1969 study systematically sampled the lower half of the formation up to the lower part of the Sewemup Member along the base of the cliffs bordering the Dolores River 15 km south of Gateway, Colorado. The localities from which these samples came are shown by the dots in Figure 1. Samples from the upper half of the formation (for the present study) were collected near the junction of Salt Creek with the Dolores River, and the sites are shown in Figure 1 by the asterisks. Sampling for the present study was begun about 33 m below the Sewemup–Ali Baba boundary and overlaps the previous work (Helsley, 1969) by approximately 75 m. Stratigraphic

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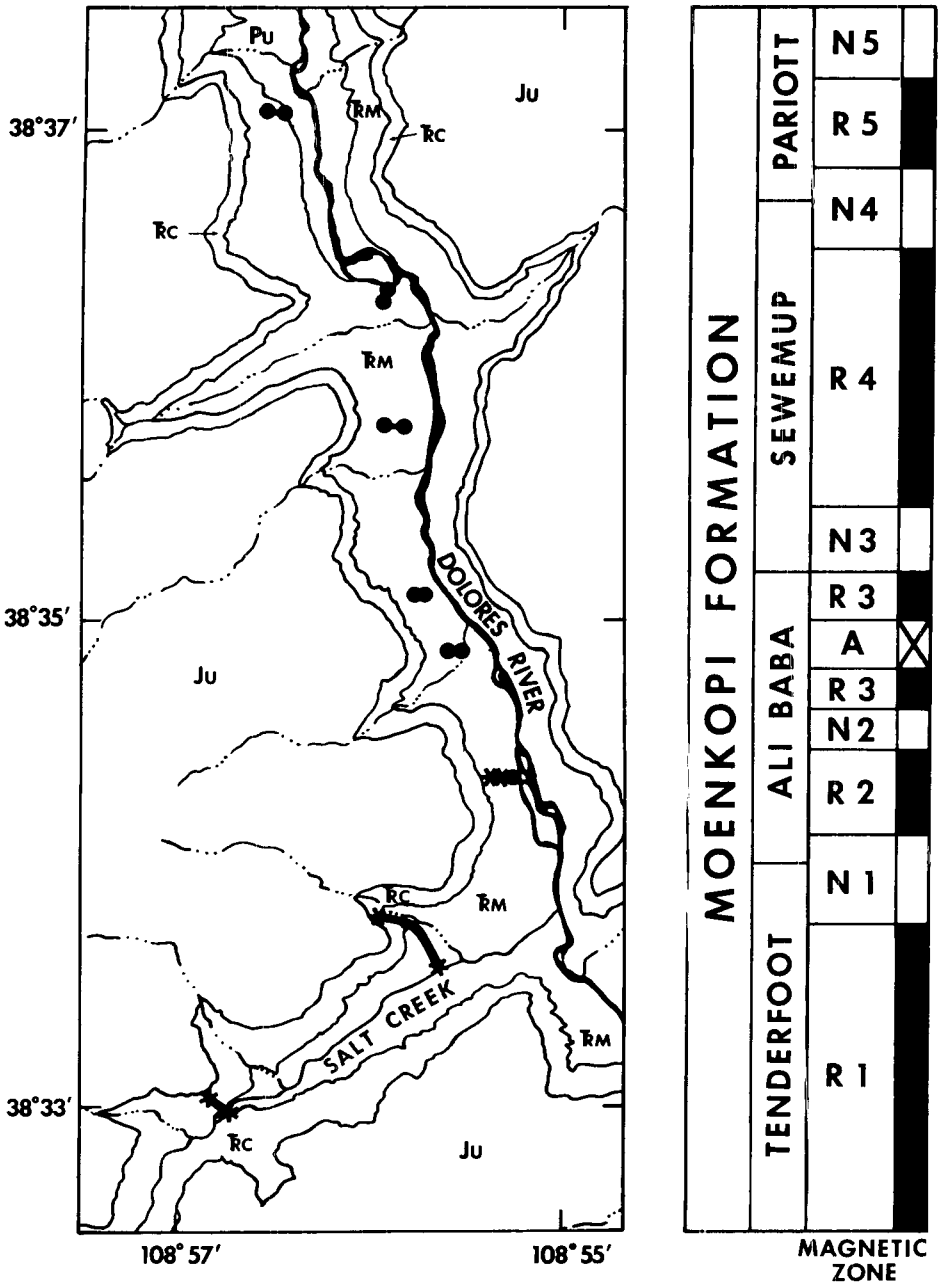


Figure 1. Location map showing geology of the collecting sites in central western Colorado and the general reversal sequence as related to the four members of the Moenkopi Formation. Pu = Permian Cutler Fm., Trm = Triassic Moenkopi Fm., Trc = Triassic Chinle Fm., Ju = undivided Jurassic Fms. Single and double dots represent Helsley's (1969) localities; asterisks indicate localities of the present study. The long continuous sample sequence in Salt Creek is indicated by broad line starred on either end.

sampling at 35-cm intervals was carried out up to the base of the overlying Chinle Formation, for a total section of 150 m. Cores were collected by means of a hand-held gasoline powered drill (Helsley, 1967) and were oriented by magnetic compass.

Three kilometers farther up Salt Creek, the basal portions of the uppermost member of the Moenkopi Formation are exposed. This unit, the Pariott Member, is present in only a few places in the western Colorado–eastern Utah area. The Pariott has tentatively been correlated with the exposures of the Moenkopi Formation in Arizona and may be of Middle (?) Triassic age (Shoemaker and Newman, 1959). At the locality in Salt Creek, 35 m of the Pariott Member and 8 m of the unconformably overlying Chinle Formation were sampled.

NATURAL REMANENT MAGNETIZATION

The field cores were cut into 2.5-cm-long cylinders and measured on a PAR SM-1 spinner magnetometer. Thermal demagnetization was done in a noninductively wound, twelve-sample capacity furnace. Temperature was monitored at four levels within the furnace by platinum thermocouples referenced to an ice-water mixture and read on a digital voltmeter. Care was taken to ensure that the samples cooled in a zero field region (± 10 gammas). After cooling from each demagnetization step, the samples were measured on the PAR magnetometer.

The natural remanent magnetization (NRM) of all samples of the upper part of the formation is summarized in Figure 2. For purposes of discussion and identification, the formation has been subdivided into polarity zones beginning at the base of the formation (Fig. 1). Sampling of this second profile of the Moenkopi Formation began in the reversed zone R3 (Helsley, 1969) and overlaps the previous study from zone $R3_A$ to the lower part of zone R4 (Figs. 1 and 4).

The detailed study of the upper part of the Moenkopi Formation reveals one thick reversed zone followed by three thinner polarity zones. Sampling was continued across the Moenkopi boundary into the overlying Chinle Formation. The uppermost Moenkopi samples measured in this study are of normal polarity (N5), and the overlying unconformable basal units of the Chinle Formation are reversed.

The NRM of the reversed groups has a mean declination of 159° and inclination of −2° and the normal intervals, 348° and 24°. The two NRM means are approximately 160° apart. The NRM directions of several portions of the upper Moenkopi Formation show large dispersion, particularly in zones $R3_B$, R5, and N5. All parts of the section showing highly scattered NRM directions have abundant coarse sandstone units, and thus the greater scatter can be correlated directly to the presence of coarse material.

R4 has a much smaller degree of scattering of NRM directions than any of the other reversed sequences. This consistent direction persists for a long stratigraphic interval (approximately 60 m) and the low degree of scatter is probably related to lithology, for this reversed zone occurs in a section of finer grained sediments in comparison to the lithology of the other reversed zones. The extreme scatter characterizing the NRM of the A and R3 zones ceases abruptly near the base of N3 and corresponds to the lithologic boundary between the coarse sands of the Ali Baba Member and the siltstones and fine sands of the Sewemup Member.

The continuation of work on the Moenkopi Formation brought to light a correlation error in Helsley's original collection. In the 1969 paper, four normal

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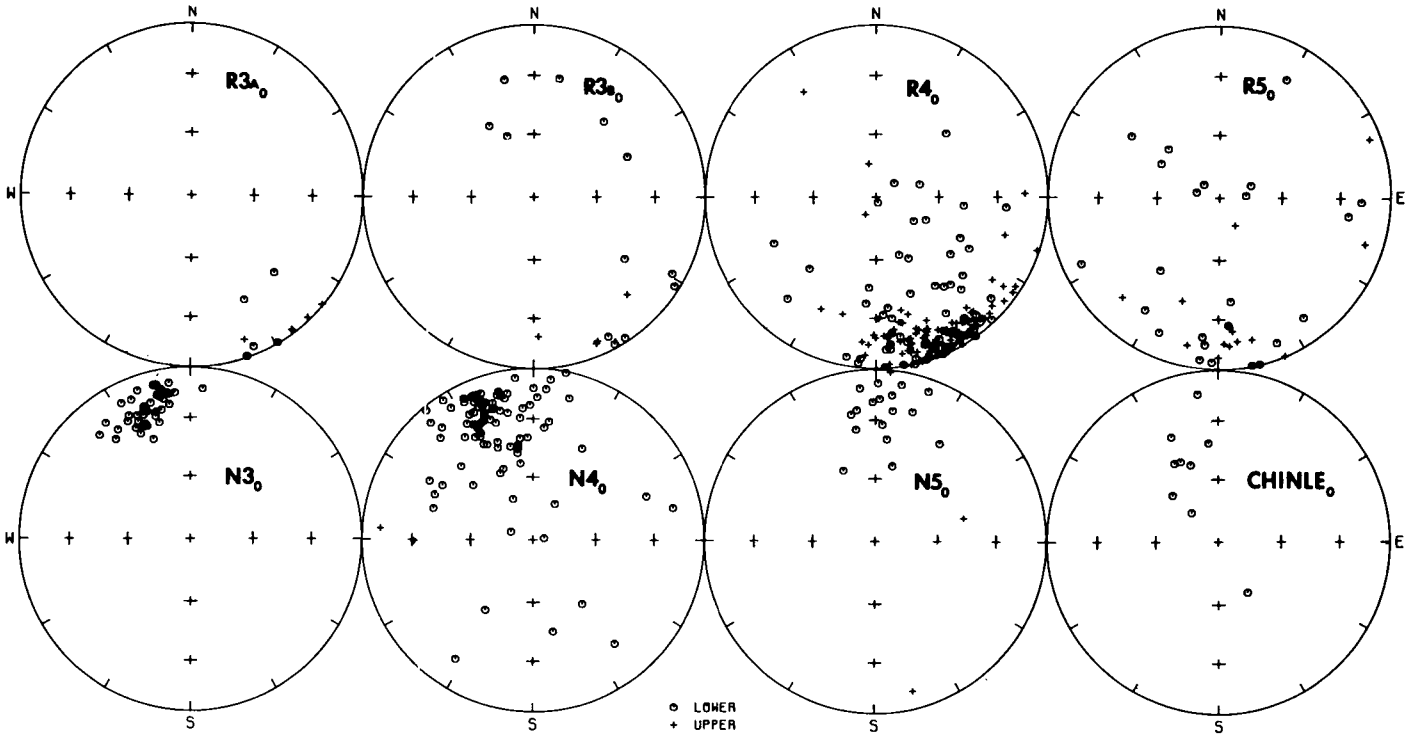


Figure 2. NRM directions in each polarity zone in the upper part of the Moenkopi Formation and in the base of the overlying Chinle.

and five reversed zones were described. It has since become apparent that the section containing polarity change N3 to R4 was sampled twice; thus N4 and R5 of Helsley (1969) are the same as N3 and R4. The lower part of the Moenkopi therefore contains three normal and four reversed zones. To verify the new correlation, an additional 12 m of section above Helsley's 1969 section were sampled, thus extending that section well into the R4 zone and permitting an unambiguous lithologic correlation to be made between Helsley's previous sections and the present study.

DEMAGNETIZATION STUDIES

Helsley (1969) demagnetized a portion of the lower Moenkopi samples in alternating fields of up to 1400 oersteds. It was found that the coercive force of the secondary component was approximately the same as that of the NRM, and thus AC demagnetization was not an effective cleaning procedure for samples from the lower part of the Moenkopi Formation. Thermal demagnetization proved to be quite effective in removing the secondary components responsible for the scatter in the samples from the lower part of the Moenkopi Formation, and thus all samples from the upper Moenkopi section were thermally demagnetized in two steps—550°C and 620°C. Some samples were also heated to 660°C. In general, the direction of magnetization does not change when the samples are heated to temperatures in excess of 550°C. These results are shown in Figures 3 and 4. The mean of the reversed samples after demagnetization is 160°, −13°; the mean for the normal samples is 342°, 17°. The two means are 176° apart, indicating that a stable magnetization has been isolated. Table 1 shows the mean direction of magnetization of each polarity zone after thermal demagnetization.

Thermal demagnetization significantly reduced the scatter of this sample population. The result is a highly consistent set of data showing well-defined polarity zones. The internal consistency of the data allows the identification of several small magnetization anomalies within the section. The "anomalous" zones constitute only about 5 percent of the samples, and this suggests that anomalous behavior should be a minor part of most sample sets. The anomalies observed are an "excursion" of directions in the R4 zone, an anomalous interval of magnetization within the R3 zone (identified as zone A), and an undulating variation of magnetization direction throughout N3-R4-N4 zones.

In the R4 zone, a group of samples, about 15 m above the base of the interval (Fig. 4), seem to show a transitional movement away from and back to the typical reversed direction. An intensity low precedes the variation in direction, a feature also observed in the succeeding reversal of polarity (R4-N4). The systematic variation of direction and intensity of magnetization observed in this interval suggest that this anomaly may be a short term excursion of the Earth's magnetic field. Its observation in a second sampling of the lower part of the R4 zone several kilometers away further substantiates its validity.

The zone referred to as PF by Helsley (1969) is again present in this section and is hereafter referred to as zone A. Zone A is present in the position expected from the previously sampled section (approximately 5 km away) and encompasses approximately the same stratigraphic thickness (Fig. 4). The NRM directions (Fig. 5) from this collection of zone A are more scattered than were the magnetization directions of the samples collected previously, and the intensity is low compared to the mean intensity of the rest of the upper Moenkopi samples (3.43 versus 7.18 x 10^{-6} emu/cc). The pole position obtained from the most consistent part of the zone is four degrees from the zone A pole obtained by Helsley (1969; see Fig. 11 below). Zone A is therefore reproducible at neighboring sections and may well be a real feature and not just a local anomaly.

In Figure 4, an undulatory behavior of the magnetization with respect to stratigraphic position is quite noticeable, particularly in zone R4. This behavior is most obvious in the part of the section consisting of N3, R4, and N4 and is restricted to samples from the Sewemup Member, a unit of fine-grained

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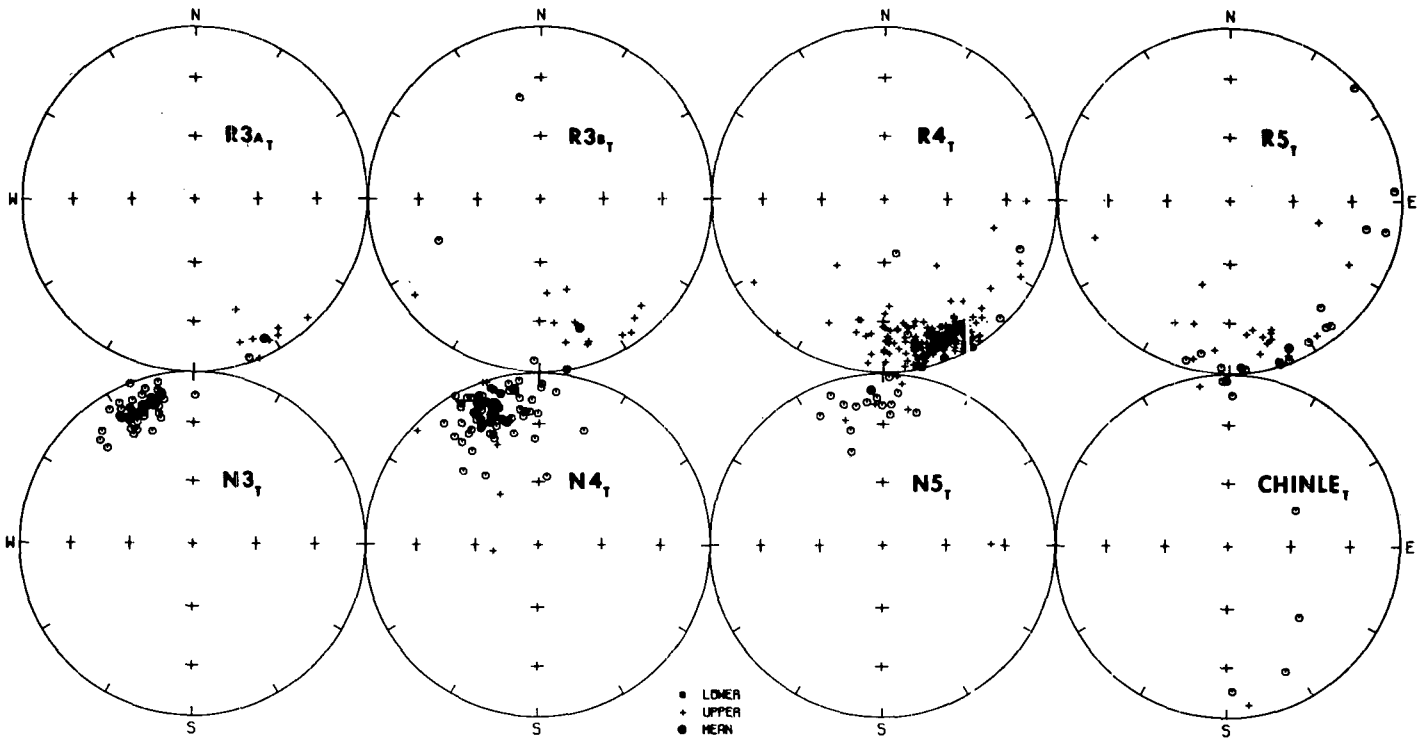


Figure 3. Directions and means of each polarity zone after thermal demagnetization to about 550°C.

lithology. In order to test the reality of this behavior, the original sample population was subdivided into two groups, and, to make these groups independent, every other sample was assigned to alternate groups. Running averages, over ten samples, were calculated on each of the two sample groups. The striking similarity of the resulting curves (Fig. 6) shows that the undulatory behavior is indeed a real feature of the remanence and not just an artifact produced by a few samples. In Figure 6 the polarity of the normal zones has been inverted so that undulation of the directions could be observed more readily. Since the polarity boundaries cannot be identified in the running average data (Fig. 6), nor can they be seen in a plot of individual sample directions converted to one polarity, one is led to the conclusion that the variation is independent of polarity reversals. This suggests that the anomalous (secular ?) field is independent of the main dipolar field.

Poles have been calculated for every 3-m stratigraphic interval for all of the upper part of the formation and are shown in Figure 7, a through g. These poles, particularly those from the N3-R4-N4 section (7, b through f), have directions that range from a Lower Permian pole position (Gose and Helsley, 1972) to a Lower Jurassic position (Steiner and Helsley, 1972). This is not one continuous change but seems to oscillate back and forth. Helsley's previous collection of the Moenkopi Formation was also analyzed in the same manner for that part of the section corresponding to R4. Although his was only the lower half of R4, the poles produced exactly the same pattern as those from this sampling of zone R4 (compare Fig. 7, b through d with 7h).

To demonstrate the range of the variation, six groups of samples, one from each of the more or less constant directions observed in Figure 6, were analyzed for a pole position. The result was either a "Permian" pole, a "Lower Triassic" pole, or an "uppermost Triassic" pole, depending on where one sampled the section. Therefore, there is a large probability that a pole position derived from a short stratigraphic section would not have been a valid representation of the pole for the entire formation. If one wishes to remove this long-term variation, one must sample the section more or less uniformly over a large stratigraphic (and presumably time) interval.

TABLE 1. ANALYSES OF DIRECTIONS OF EACH POLARITY INTERVAL

	D	I	N	R	K	α_{95}	Long	Lat	dp	dm
All Data										
$R3_A$	153.2	-9.9	9	8.79	39.0	8.3	113.7	48.6	4.3	8.4
$R3_B$	151.6	-21.9	16	13.69	6.5	15.7	107.1	58.3	8.8	16.6
N3	338.3	16.7	44	43.29	60.5	2.8	109.6	54.2	1.5	2.9
R4	160.0	-13.3	171	161.36	17.6	2.6	105.3	53.5	1.4	2.7
N4	340.9	18.1	80	75.05	16.0	4.1	106.4	56.1	2.2	4.2
R5	158.3	-9.3	35	27.97	4.8	12.4	106.8	50.9	6.3	12.5
N5	355.9	10.5	19	17.83	15.4	8.8	78.5	56.6	4.5	9.0
CHINLE	176.3	-11.5	9	7.60	5.7	23.6	77.8	57.1	12.1	23.9
Selected Data										
$R3_A$	153.2	-9.9	9	8.79	39.0	8.3	113.7	48.6	4.3	8.4
$R3_B$	156.9	-20.2	9	8.66	23.4	10.9	113.6	55.1	6.0	11.4
N3	338.8	16.4	43	42.36	65.3	2.7	108.9	54.3	1.4	2.8
R4	160.3	-13.0	156	151.21	32.3	2.0	105.2	53.5	1.0	2.0
N4	341.7	16.9	72	69.73	31.3	3.0	104.6	55.9	1.6	3.1
R5	165.5	-8.1	23	21.92	20.4	6.9	95.5	53.2	3.5	6.9
N5	353.5	17.5	14	13.57	30.3	7.3	83.9	59.9	3.9	7.6
CHINLE	175.2	-3.8	5	4.96	94.4	7.9	77.3	53.3	4.0	7.9

Longitude is east; latitude is north.
N = Number of samples or zones.
R = Resultant vector.
K = Precision parameter.
α_{95} = Semiangle of 95% confidence cone.
dp, dm = Semiaxis of error ellipse.

INTENSITY

The mean NRM intensity of magnetization for the upper part of the Moenkopi Formation is 7.8 x 10^{-6} emu per cc, and intensities range from 0.5 to 35.0 x 10^{-6} emu per cc. The intensity after thermal demagnetization to the tightest sample grouping (550°C for 85 percent of the samples and 630°C for the remainder) is 6.20 x 10^{-6} emu per cc. The intensity after demagnetization for this part of the formation is approxi-

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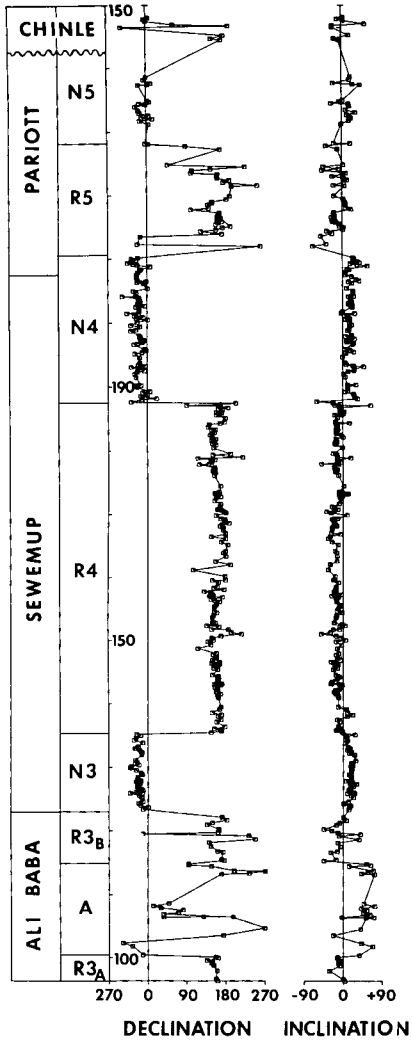
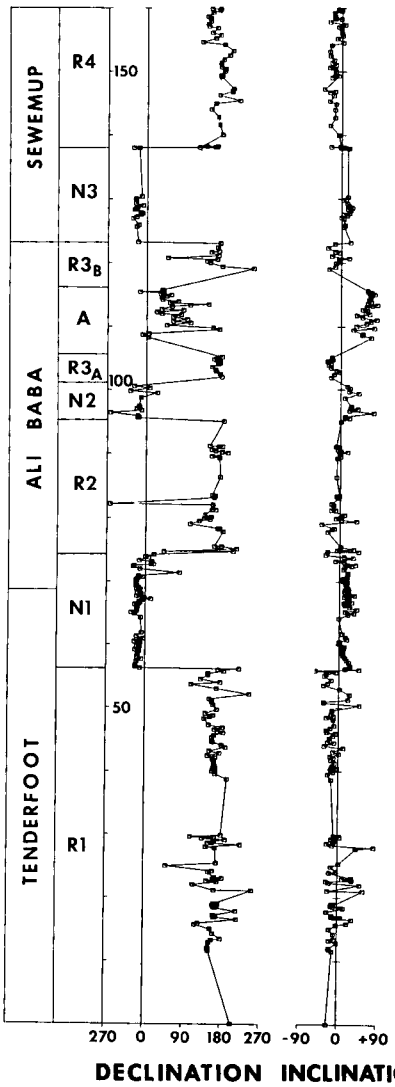


Figure 4. Stratigraphic plot of sample declinations and inclinations after about 550°C demagnetization for the entire Meonkopi Formation. The stratigraphic sequence on the right is from Helsley (1969).



mately the same, though slightly higher, than that for the lower part of the Moenkopi Formation (4.56×10^{-6} emu per cc; Helsley, 1969). Figure 8 is a histogram of sample intensities, grouped according to normal or reversed polarity, both before and after thermal demagnetization. As was seen in the lower part of the Moenkopi Formation, the intensity of the normal polarities is virtually the same before and after thermal demagnetization as are those of the reversed polarities. And, as in the lower part of the Moenkopi Formation, the normal and reversed intensities are not the same before demagnetization. The mean normal intensity is higher by 0.87×10^{-6} emu per cc, but is only 0.36×10^{-6} emu per cc higher after demagnetization.

The mean intensities of each polarity zone are shown in Figure 9. Helsley (1969) noted an increase in intensity upward in the lower part of the Moenkopi Formation. In contrast, there seems to be a decrease upward in the upper part of the Moenkopi Formation. The combination of these data result in the pattern seen in Figure 9 for the entire Moenkopi Formation . This variation parallels the variation of lithology and indicates that the finer grained lithologies of the Sewemup and Ali Baba Members have relatively high intensities, but that the coarser sediments of the Pariott, Ali Baba, and Tenderfoot Members have low intensities.

POLES

No standard method of analysis has been adopted for computing a pole position for a lengthy stratigraphic interval with a large number of samples. One may analyze all samples together; but for the sake of comparison, it is obvious that such a pole will represent a tremendously greater time span than, for example, a pole from one or several lava flows. Thus, it is felt that it is more meaningful to compute poles by intervals throughout a long stratigraphic section. It is reasonable to examine intervals of different polarity or intervals of an arbitrary stratigraphic length.

Polarity interval analysis is advantageous in that changes in polarity represent natural divisions of the magnetic record. However, there is the disadvantage of computing pole positions from large and nonuniform units of time per pole. A more uniform analysis can be obtained by analyzing the observations in terms of a fixed stratigraphic interval whose length is determined by the shortest polarity interval observed, provided the density of sampling is high enough. The shortest polarity interval recorded in this study is 6 m. For this reason, the observations were analyzed in 6-m intervals. This interval per pole (15 samples per pole) proves to be sufficient to average out lithologic variation and samples of uncharacteristic magnetization. The precision of the pole as reflected in the oval of confidence (dp, dm) is quite acceptable for the 6-m groups, averaging about 6° to 8° for the long axis (dm).

However, the 6-m interval tends to average out much of the long period variation noted above, and so a shorter interval of 3 m was also used. Analyzing by 3-m intervals with the sampling density used in this study gives an average of about eight samples per pole calculation. Analysis by 3-m intervals (Fig. 7, a through g) reflects the

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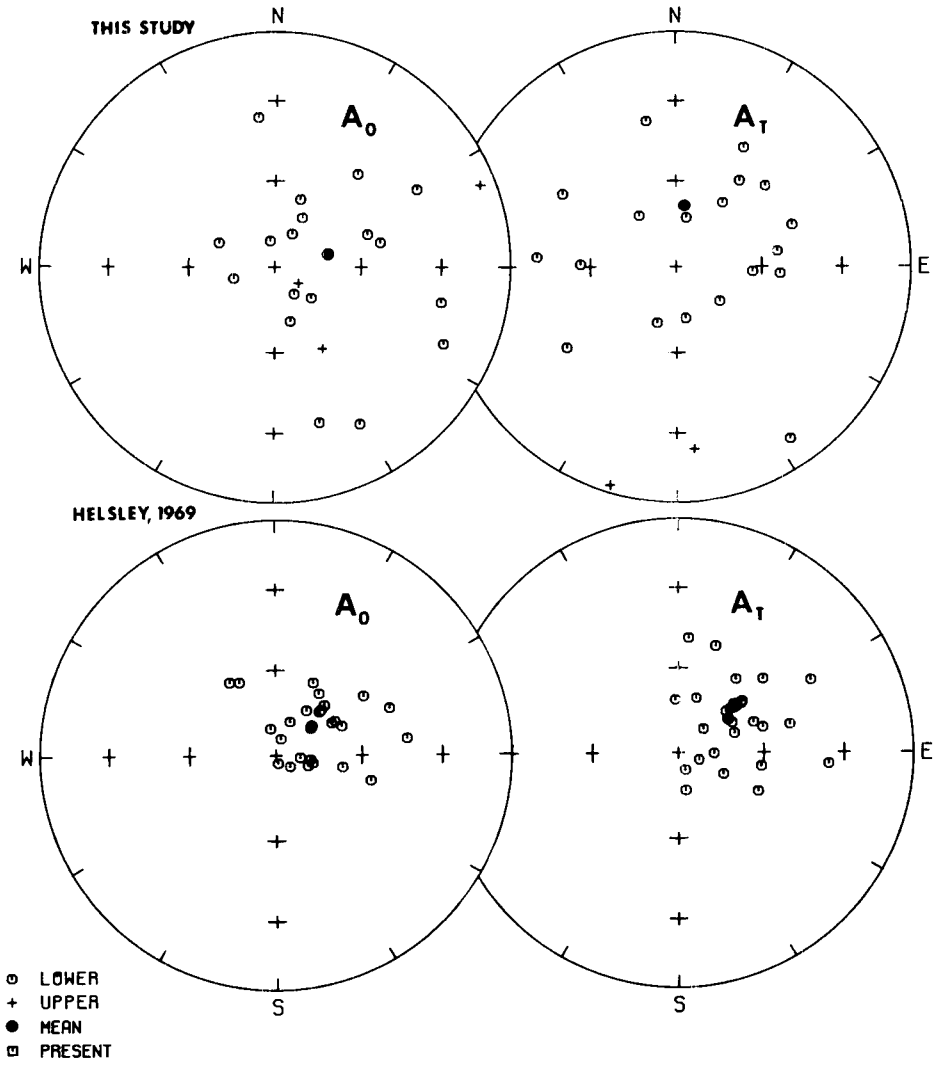


Figure 5. Anomalous zone NRM (A_0) directions and directions after about 600°C (A_T) from both the present study and Helsley's 1969 study.

long-term variations of the pole seen in N3-R4-N4 and may be the best way of analyzing this aspect of the data.

All analyses are complicated by the problem that a few samples have directions far removed from the means of their respective groups. These can perturb the mean of a small group significantly. In a large collection, these abnormal directions could quite reasonably stem from orientation errors in the field, from errors in recording the dip and strike of the core in the field, or from sampling a lithologic unit unsuitable for paleomagnetic study (for example, a coarse sand lens). As experimental and natural errors cannot be ruled out, it does not seem unreasonable to exclude the very abnormal samples from any group. Thus, one can calculate the above four analyses of poles with all data present (unselected) or with abnormal samples removed (selected). Demagnetized data are preferred over NRM data according to theory, but an NRM pole analysis was also computed. (Due to the presence of a secondary magnetization in the NRM data, the distinction between added secondary directions and aberrant directions is not possible, and no selection was made.)

The results of these different types of analyses in both the selected and unselected forms are presented in Table 2. An over-all pole was calculated from the group means for each type of analysis. Figure 10 shows that all methods give almost exactly the same pole. If one considers the question of statistical accuracy of the pole position (Table 2), the analysis of 6-m intervals seems to give the best combination of a small circle of confidence (α_{95}) and a high precision parameter (k). The 6-m interval poles average out most of the long-term variations observed and are used as the estimate of the pole for this sequence of sediments.

For the analysis of the upper part of the Moenkopi Formation, however, the question of the best analysis is primarily academic, as even the NRM pole position is within 4° of the variously computed demagnetized poles. The agreement between the NRM and demagnetized poles is at first surprising, considering the amount of instability shown in Figure 2. The agreement results from the method of analysis, which was to invert any data point that was reversed before adding it to the accumulating mean. For the points in the streak between the reversed direction and the present Earth's field, this meant that the directions more than 90° from the mean normal direction were inverted while those closer than 90° to the normal direction were left unchanged. The result was to artificially balance the streaked points on either side of the true normal direction. A good agreement between the NRM and demagnetized data can always be expected for a set of samples collected from a mixed polarity sequence whose unstable samples are more or less evenly distributed along the instability great circle, provided some stable samples are present to determine the boundary between the data to be left unchanged and that to be inverted.

The pole position (6-m interval, selected)

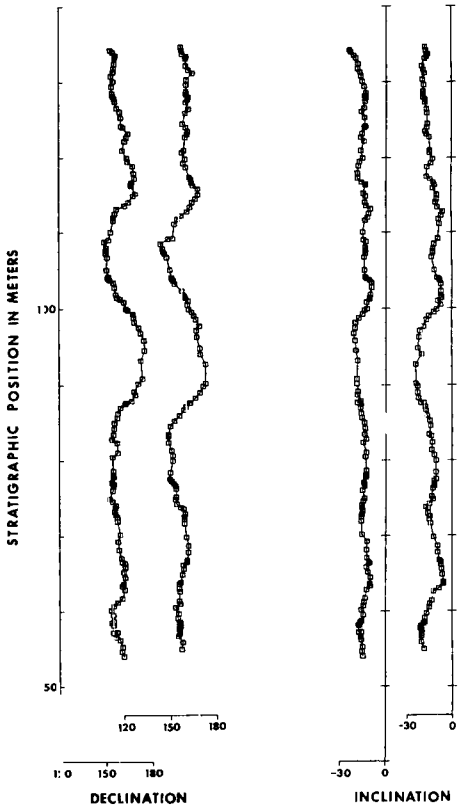


Figure 6. Running averages over ten samples for two independent samplings of the N3-R4-N4 interval obtained by assigning every other sample to alternate groups. Directions used are from samples demagnetized to approximately 550°C.

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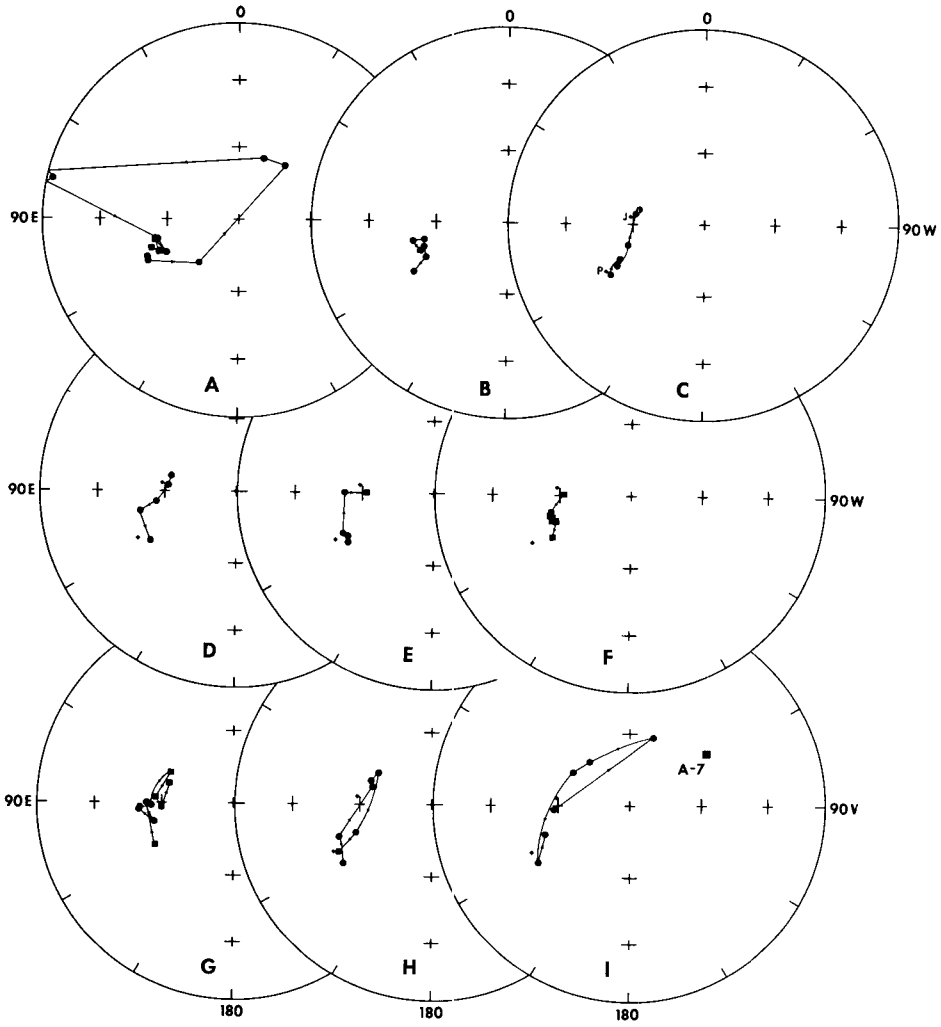


Figure 7. a through g. Three-meter-interval poles from the upper Moenkopi Formation. h. Three-meter-interval poles for the lower half of R4 in Helsley's 1969 collection. i. Pole for each sample of the excursion in R4 and the pole for anomalous zone A. Gose and Helsley's (1972) Permian pole (P) and Steiner and Helsley's (1972) Lower Jurassic pole (J) are plotted for reference as small diamonds.

for the upper Moenkopi (103° E., 55° N.; dp 1.6°, dm 3.2°) compares well with the one from the lower Moenkopi (98° E., 57° N.; dp 2.5°, dm 4.9°) as would be expected. The slightly more northerly position of the pole from the lower part of the formation is probably artificial, as the normal samples contributing to it were, for the most part, not demagnetized. The upper Moenkopi Formation pole is compared to the lower Moenkopi Formation pole (Helsley, 1969) and other published Lower Triassic poles in Figure 11. The upper and lower Moenkopi Formation poles are very close together and, in general, agree well with the other observations of Triassic pole positions (summarized by Helsley, 1969, Table 8).

DISCUSSION OF RESULTS

In the course of sampling the higher part of the Moenkopi Formation, a certain amount of the lower part of the formation was resampled. This dual sampling is interesting because it allows correlation of part of the polarity sequence over a distance of several kilometers. The overlap of the sections include the zones $R3_A$, A, $R3_B$, N3, and the lower part of R4. The same reversal sequence is present in both, and the stratigraphic lengths of the polarity zones are approximately the same. Moreover, the long-term variation of the pole observed in R4, the "anomalous" zone A, and the "excursion" in the lower part of R4 are present in both sections and suggest that small-scale irregularities may be very useful in separating one polarity zone from another when they are used in regional correlation.

Two polarity zones are strikingly different from the others in the Moenkopi Formation and may be useful in regional correlation within the Moenkopi Formation and within other time-equivalent units. They are the reversed zones R1 (including R1S) and R4, each of which is about 60 m long. Unfortunately, R1 is incomplete because its base is an unconformity. Polarity zone R4 covers a lengthy stratigraphic interval between thinner zones of normal and reversed polarities. In addition to its greater thickness (and presumably longer time duration), R4 has recorded a systematic undulatory variation throughout its extent and a small excursion in its lower part. These features can be correlated for several kilometers at least, and thus the R4 zone seems to present a very useful tool for correlation within the Lower Triassic.

The Moenkopi Formation and, if its age designations are correct, the Lower Triassic and possibly the lowest part of the Middle Triassic are characterized by at least nine polarity changes of the geomagnetic field. In view of the facts that the short polarity zones of the Cenozoic follow a long constant polarity period in the Cretaceous (Helsley and Steiner, 1969), and the short Triassic polarity zones follow the long constant

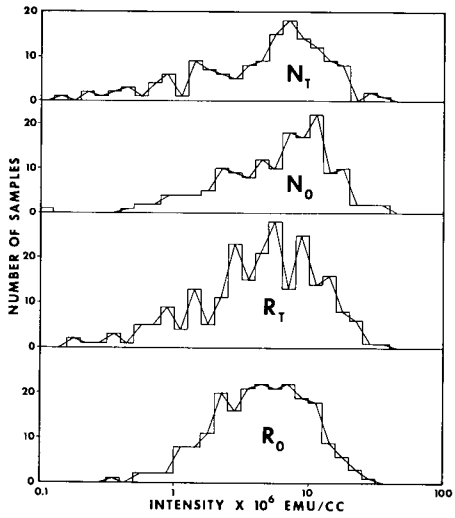


Figure 8. Intensity histograms of normal (N) and reversed (R) NRM (O) and demagnetized (T) data.

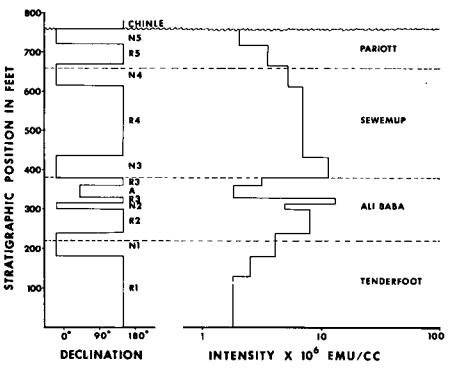


Figure 9. Average intensities of each polarity interval (indicated by a schematic declination plot) plotted according to stratigraphic position. Members of the Moenkopi Formation are indicated on the right.

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TABLE 2. ANALYSES OF UPPER MOENKOPI FORMATION SAMPLE DIRECTIONS

	D	I	N	R	K	α_{95}	Long	Lat	dp	dm
All samples NRM	160.0	-12.3	387	322.56	6.0	3.2	105.6	53.0	1.7	3.3
All samples demag. (un)	160.9	-14.7	374	346.78	13.7	2.0	105.0	54.5	1.1	2.1
All samples demag. (sel)	161.0	-14.3	326	315.38	30.6	1.4	104.6	54.4	0.7	1.5
By polarity zones (un)	161.2	-14.3	7	6.94	93.6	6.3	104.4	54.4	3.3	6.4
By polarity zones (sel)	161.4	-14.7	7	6.95	110.7	5.8	104.2	54.7	3.0	5.9
By 6-m intervals (un)	162.1	-14.8	22	21.78	96.9	3.2	103.2	55.0	1.7	3.2
By 6-m intervals (sel)	162.0	-14.5	22	21.79	101.9	3.1	103.3	54.8	1.6	3.2
By 3-m intervals (un)	162.1	-14.5	43	41.81	35.2	3.7	103.1	54.8	2.0	3.8
By 3-m intervals (sel)	162.4	-14.5	43	42.40	70.0	2.6	102.6	54.9	1.4	2.7
A zone (un)	8.3	69.3	21	15.67	3.8	19.1	90.1	74.4	27.8	32.6
A zone (sel) A-7	53.0	68.4	14	12.49	8.6	14.4	56.6	51.3	20.4	24.2
A zone 1969	46.7	66.9	28	26.59	19.2	6.4	52.9	55.3	8.7	10.6

See Table 1 for explanation of heading abbreviations.
Un = unselected (all samples are in analysis).
Sel = selected (deviant samples removed).
Demag. = thermal demagnetization.

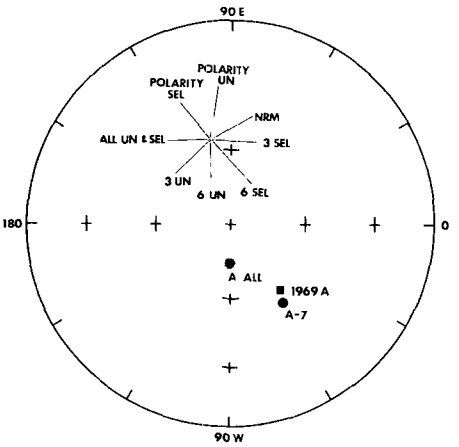


Figure 10. Poles (dots at center of spoke) computed by various analyses (polarity zones, 3-m intervals, and 6-m intervals) for the entire Moenkopi Formation. SEL = selected data, UN = all data used. Anomalous zone poles for selected (A-7) and all data and the 1969 study are also shown.

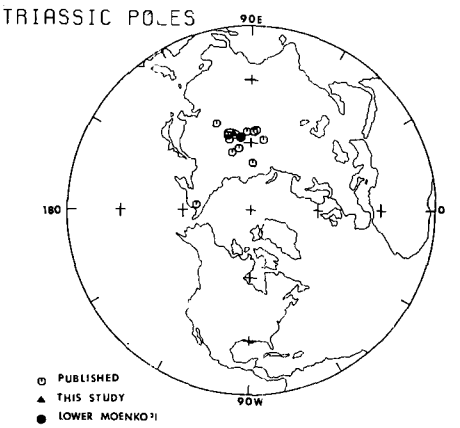


Figure 11. Upper Moenkopi pole position shown with the lower Moenkopi pole (Helsley, 1969) and published Lower Triassic poles for North America (summarized in Table 8 of Helsley, 1969).

polarity of the Upper Paleozoic Reversed Interval, it is tempting to speculate that major geologic eras, defined on the basis of major changes in the fossil record, may end with a very low frequency of reversals and begin with a high reversal frequency. The major changes in life forms that take place at these times may be the result of evolutionary adaptation during the long period of constant polarity in which the magnetic field is used in the regulation of some necessary or vital function. Such organisms would be at a severe disadvantage when a period of frequent reversals once again began; for at each reversal, the intensity of the field has been observed to decrease to a low or near-zero value. Such a speculation is supported by recent experiments in which profound biologic changes were observed during embryonic development when the organism was grown in a region of near-zero field (Russell, 1972).

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